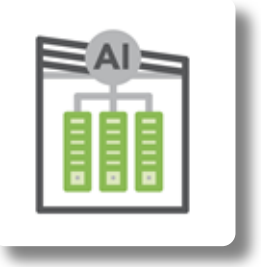
AI OPEN
DATA CENTER



Superconducting Power Distribution for AI Data Centers: Quantified Benefits and Open Challenges Applied to a 250 MW Reference Architecture

Peter Gross *PMG Associates Consulting and Advisory*

Chris Thompson *Vice President, Advanced Technology, Vertiv*

Frances Bosman *Manager – Solutions Engineering, VEIR*

OCP PLATINUM® Member

Panel Discussion



Peter Gross

PMG Associates Consulting and Advisory



Chris Thompson

Vice President, Advanced Technology, Vertiv

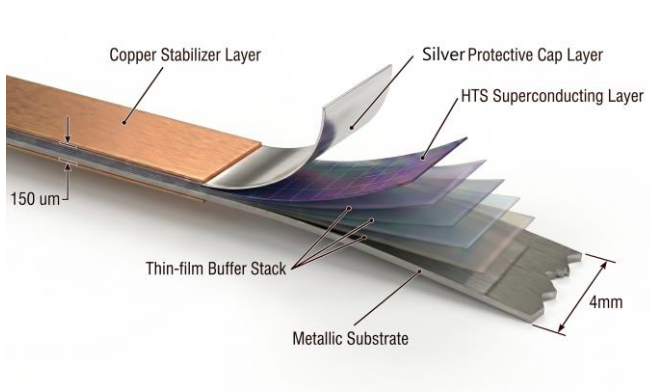


Frances Bosman

Manager - Solutions Engineering, VEIR

Superconducting Power Delivery for Data Centers

- Superconductors transfer power at higher efficiency at significantly higher ampacity.



Superconductor tape current density: ~180 A/mm2

Zero DC resistance below critical temperature (~77K for HTS).

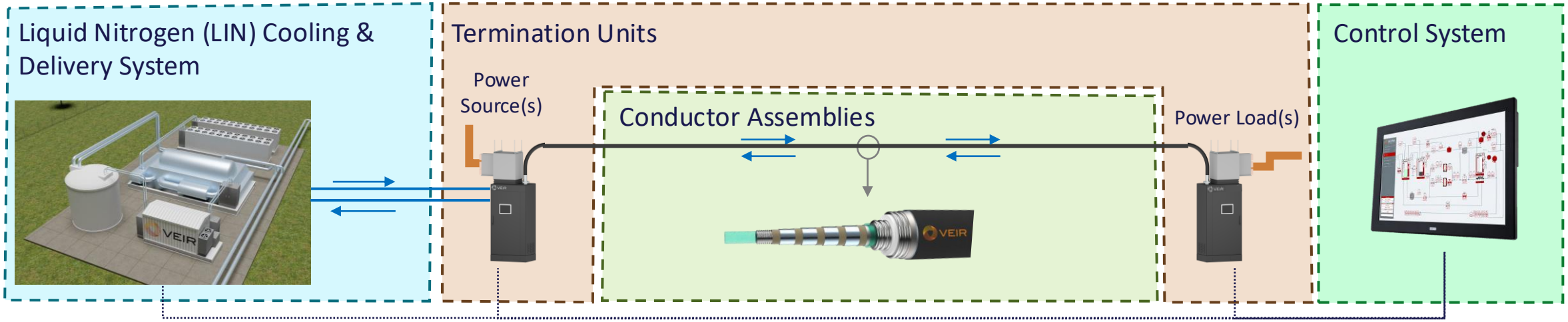
Negligible resistive losses vs. copper.



Low and Medium Voltage Solutions

- ≤ 800Vdc systems at 10 / 20 / 30 MW per cable
- ≤ 600Vac systems at 5 / 10 MW per cable
- Medium voltage systems up to 36kV at 240 MW per cable
- Single and multi core cable arrangements

System Architecture



Direct replacement of copper in 250MW Design

MV RMU loops

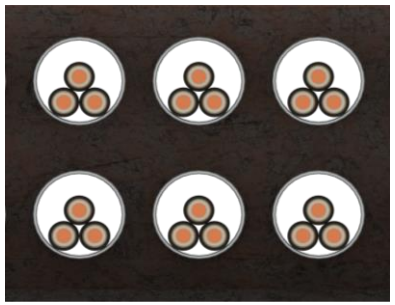
One HTS cable per MV feeder

Metric	HTS	Copper
Solution	1 cable	Parallel conds.
Copper required	1.8 kg/m	65.5 kg/m
Excavation vol.	312 yd³	1820 yd³
Concrete vol.	164 t	644 t

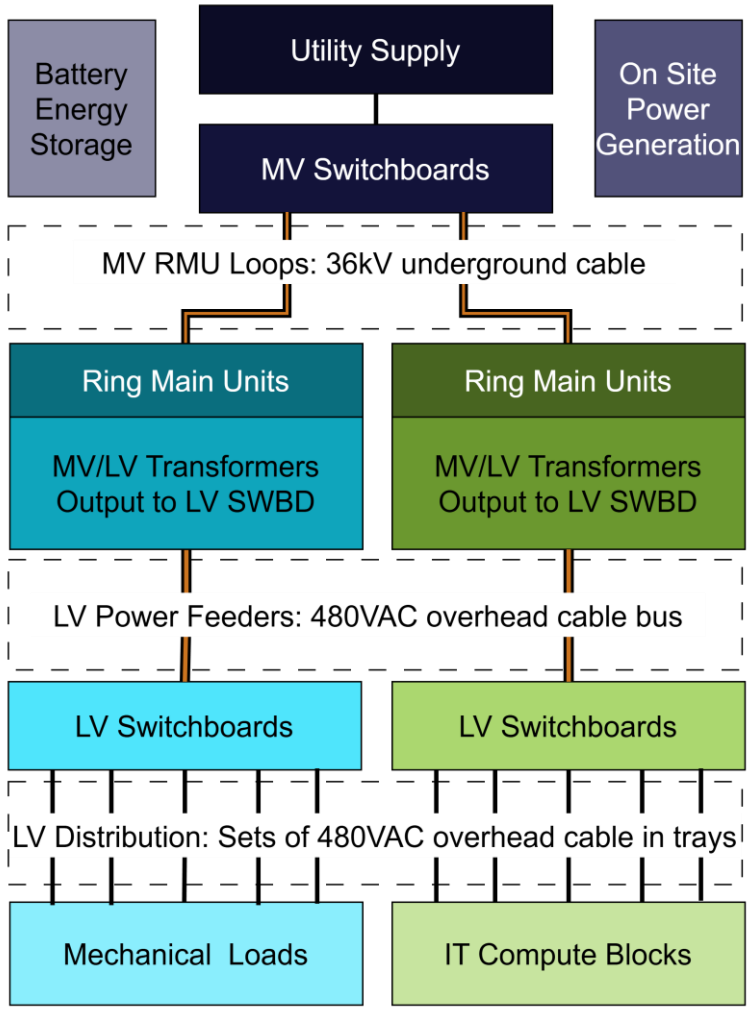


4000A ST Cable

Direct buried with concrete cap



MV Power feeder ductbank



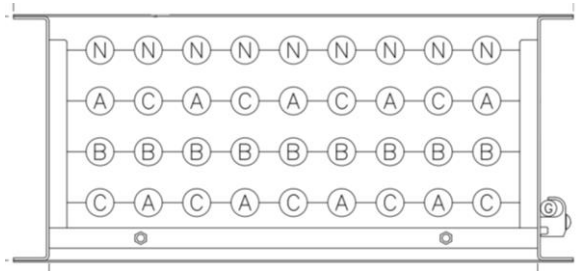
LV overhead power feeder

480 VAC, triaxial (5,000 A) cable

Metric	HTS	Copper
Solution	1 cable	Cable bus
Section length	Up to 200m	Up to 7.5m segments
Weight	12.3 kg/m	156 kg/m
Envelope	110 mm OD	700×300 mm



4000A SL-AC Cable



4000A Cable bus

Direct replacement of copper in 250MW Design

COPPER CABLES (5,000 Amps)

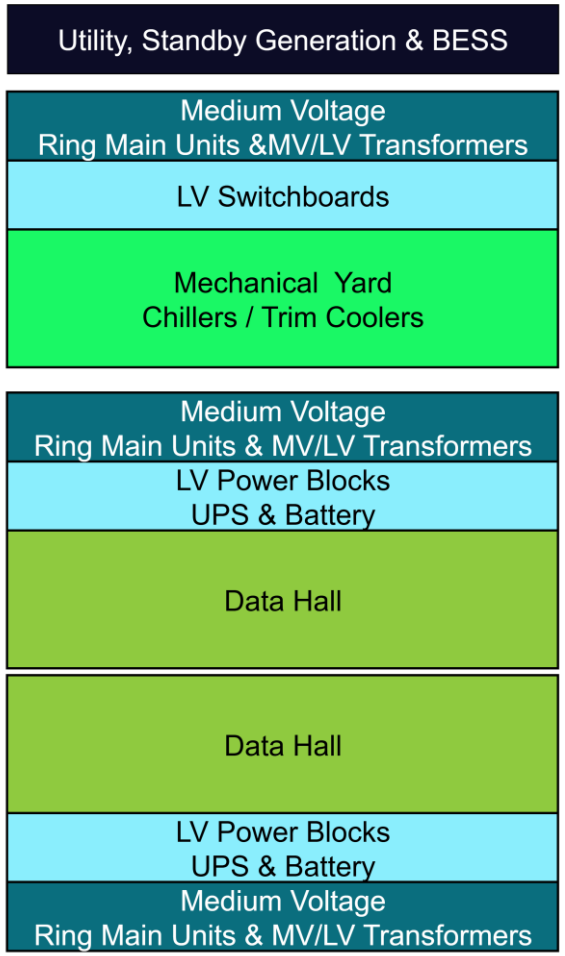


VEIR SUPERCONDUCTING CABLES (5,000 Amps)



Facility Layout Optimization

REFERENCE FACILITY LAYOUT

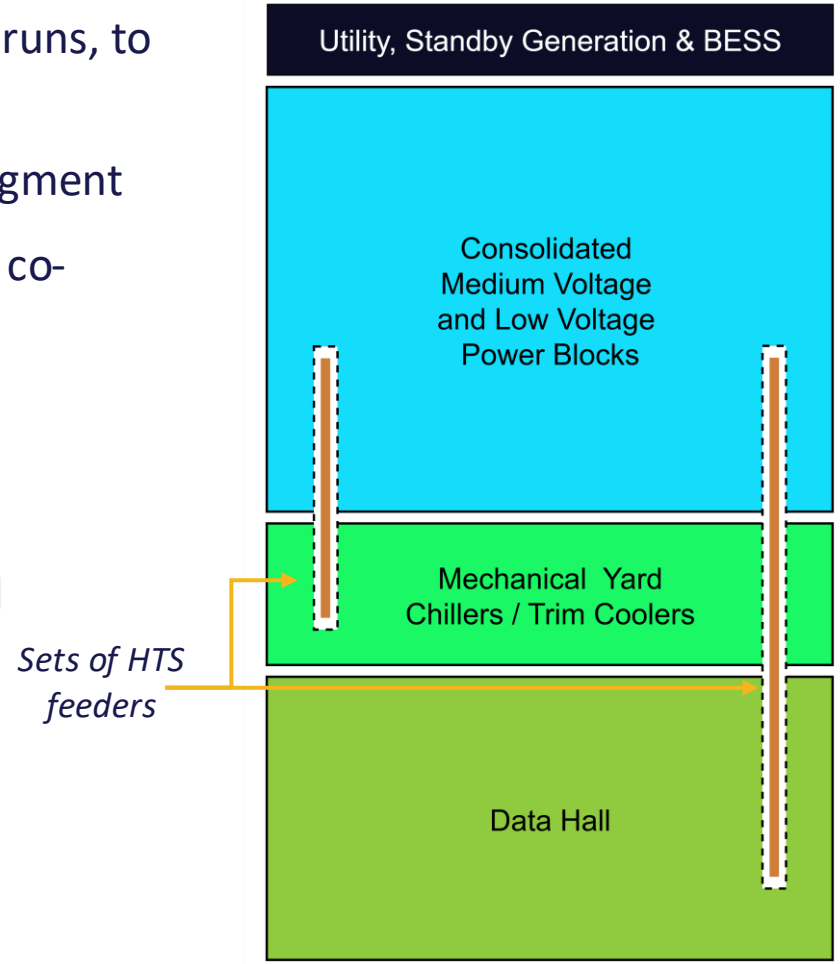


- Traditional layouts are optimized for short feeder runs, to limit voltage drop and copper volume
- HTS cables extend LV distribution to 200 m per segment
- Grey space and white space no longer need to be co-located

Layout changes enable:

- Denser IT halls & consolidated grey-space power
- Power conversion consolidated away from IT hall
- More space near racks for cooling equipment
- 25–35% reduction in water piping length
- Lower material cost and faster installation

OPTIMIZED FACILITY LAYOUT



Open Challenges for the OCP Community

- Four categories of open work stand between today's pre-commercial systems and scaled deployment.

1

Standards & codes

Cryogenic feeders, quench protection, LIN handling

2

Terminations

Integration in data center architecture

3

Fault & quench management

Cable's thermal limits and coordination.

4

Market readiness

Supply chain, field service, installer training

Call to Action

Work with the OCP community to develop open interface definitions, standards, and shared reference designs for superconducting power solutions.

OCP HTS White Paper Workstream

OCP Power Distribution Project

VEIR HTS Solutions

Open Discussion

